15/09/20

SOLID STATE

-: By ALK sir

→ Solid is characterised by definite volume, sharp & due to strong intermolecular forces.

→ Solids → Crystalline solids
→ Solids → Non-Crystalline or Amorphous or pseudo solids or Super Cooled liquids

-: Crystalline solids

1. Constituent particles are arranged in regular fashion & such solids are said to have long range order

eg:- Diamond, graphite, metals (Na, Fe, Ag etc) NaCl, $CaCO_3$, K_2SO_4 etc. ice, dry ice, solid Ar etc.

2. Have sharp or definite melting point.

-: Amorphous solids

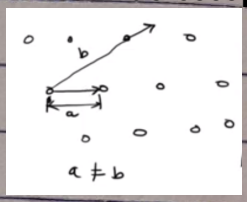
1. Constituents atoms/molecules are randomly arranged and no long range order. Such solids may have short range order.

eg:- glass, rubber, plastics.

2. Infact amorphous solids are also called as Super Cooled liquids because particles of such solids have a tendency to flow (a very slow). like liquids.

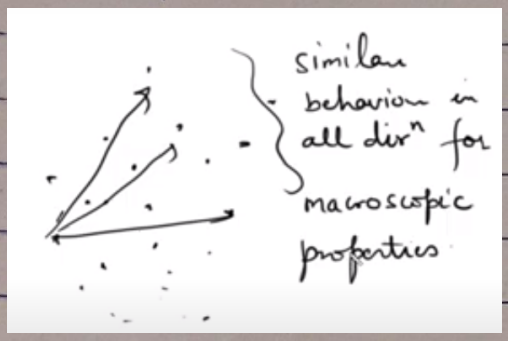
eg:- glass in window panels have greater thick

3. Called anisotropic i.e, their physical properties (like electrical Conductance, & refractive index) have different value in different directions. Macroscopic Properties



2. They melt ^ or soften over a range of temperature. (no sharp melting point)

3. Isotropic



Note:- Crystalline Solids have clevage properties. Such solids have cleavage (or weak) planes along which solids can be cut.

* Types of Crystalline solids :-

1. Metallic solids :-
   - metals (eg:- Ag, Au, Na, Fe etc)
   - Metallic bonding
   - Moderate to high M.P & B.P
   - Moderate to high hardness.

2. Ionic solids :-
   - Ionic Compounds
   - Ionic bonds
   - hard & Brittle
   - high M.P & B.P
   - eg :- NaCl, KCl, $CaCO_3$ etc.

3. Covalently bonded network solids :-

- All bonds are Covalent bonds

eg:- diamond, graphite, AlN, SiO_2, Si, Ge, SiC etc

- Very high M.P & B.P

4. Molecular solids :-

- Weak intermolecular forces {like H-bonding, dipole - dipole attr. london forces}.

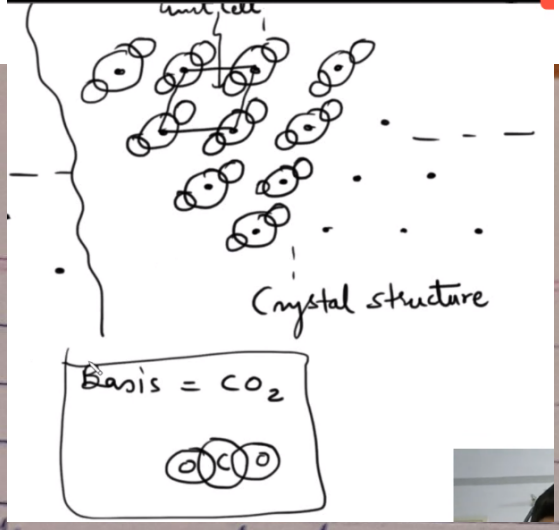
eg:- dry ice (solid CO_2), ice (H_2O) solid Ar, solid NH_3 white P_4, I_2 (solid) etc.

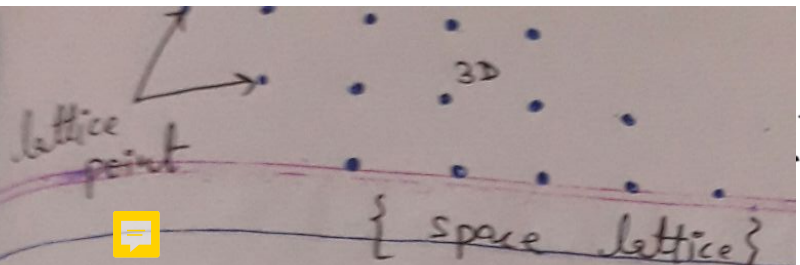
. Very low M.P and B.P

(*2) → Some definition in solid state :-

a. **Lattice (or space lattice)** :- It is a hypothetical arrangement (or array) of point in 3-D space.

b. **Basis** :- Basis is an atom or molecule or ion or a group of atoms or molecules which when placed on lattice points form crystal structure.





:- Unit cell :- It is the smallest portion of crystal lattice which when repeated in all direction produce crystal structure.

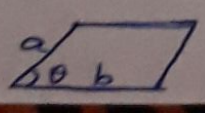
→ It is a parallelogram in (2D) & a parallelopiped in 3D which when repeated in various direction produce crystal structure.

→ Unit cell represents the entire crystal structure. if no defects in crystal, then properties of solid = properties of unit cell.

eg:- density of crystal = density of unit cell.

* Unit cells in 2D :- There are 5 different types of unit cells possible in 2D.

→ Unit cell in 2D is parallelogram which is described by 3 parameters.



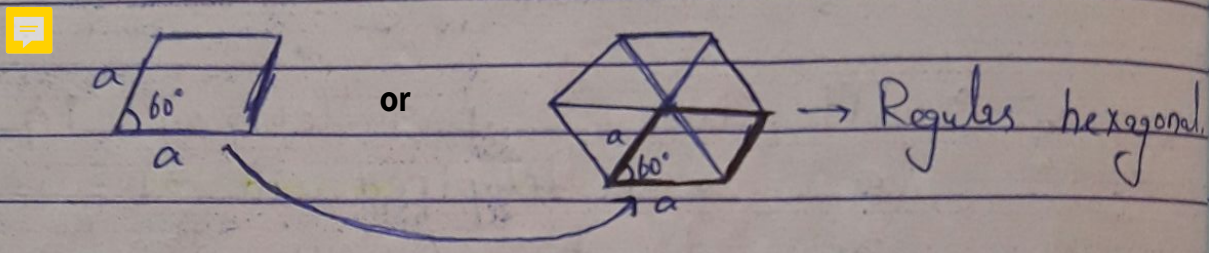
(a, b, θ)

* Types of Unit Cells in 2D :-

1. Square :- $a = b$ & $\theta = 90°$ (most symmetrical form)

2. Rectangle :- $a \neq b$, $\theta = 90°$

3. Hexagonal :- $a = b$, $\theta = 60°$ or $\theta = 120°$



4. Rhombus :- $a = b$, $\theta \neq 90°$ & $60°$ & $120°$

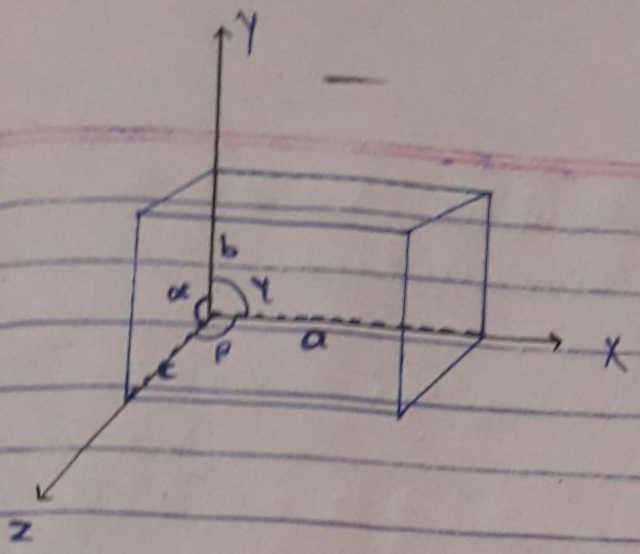
5. Parallelogram :- $a \neq b$, $\theta \neq 90°$

* Units Cell in 3D :- It is described by 6 parameters. (parallelopiped)

Three edges length & 3-inter facial angles.

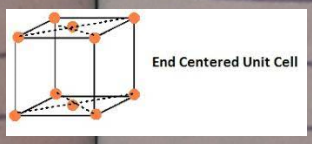
$\{\alpha, \beta, \gamma, a, b, c\}$

→ Crystallographic parameters.

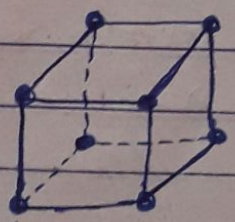


* Based on geometrical Considerations, there are 14 different types of Unit Cells possible in 3D space, Called Bravais Lattices. These 14 unit cells are grouped in 7 Crystal Systems depending upon 7 types of primitive Unit cells.

• Primitive Unit cell :- The Unit cell in which lattic points occur only at 8 Corners.
↓
most simple



eg :- In Cubic system



(Primitive)

Seven Crystal System : (NCERT)

① Cubic :-
- Primitive
- body Centered Cubic (bcc)
- face Centred Cubic (fcc)

$a = b = c$

$\alpha = \beta = \gamma = 90°$

eg :- NaCl, Zinc blende, Cu etc.

(2) Tetragonal :- • Primitive
• Body Centred

$a = b \neq c$
$\alpha = \beta = \gamma = 90^\circ$

eg:- SnO_2, TiO_2 white Sn etc. $CaSO_4$.

(3) Orthorhombic :-

$a \neq b \neq c$
$\alpha = \beta = \gamma = 90^\circ$

• Primitive
• Body Centred
• Face Centred
• End Centred

eg:- Rhombic sulphus, $BaSO_4$ etc. KNO_3

(4) Hexagonal :- • Primitive

$a = b \neq c$
$\alpha = \beta = 90^\circ$ & $\gamma = 120^\circ$

eg:- graphite, etc. CdS,

(5) Rhombohedral or Trigonal :- • Primitive

eg:- Calcite ($CaCO_3$) HgS (cinnabar)

$a = b = c$
$\alpha = \beta = \gamma \neq 90^\circ$

⑥ Monoclinic :- • Primitive
• End Centred

$a \neq b \neq c$

$\alpha = \gamma = 90°$

$\beta \neq 90°$

eg:- Monoclinic sulphus, $Na_2SO_4 \cdot 10H_2O$ etc

Glauber's salt

⑦ Triclinic :- • Primitive

most unsymmetrical form

$a \neq b \neq c$

$\alpha \neq \beta \neq \gamma \neq 90°$

eg:- $CuSO_4 \cdot 5H_2O$

$K_2Cr_2O_7$, H_3BO_3

Total unit cell = 14

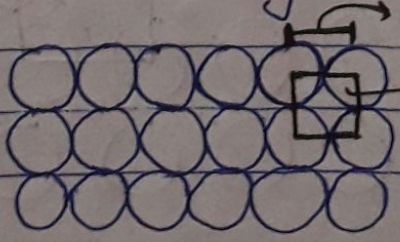
↑ Bravais lattices.

(*3) → Metallic Solids (Paking of identical spheres)

* Paking of identical sphere in 2-D :-

a. Square Packing

i) with touching spheres.



Square unit length

r → radius of sphere.

⊗ Co-ordination of Number (C.No.) = No. of nearest neighbours.

{spheres that touch} = 4

⊗ Area effeciency of such packing = $\frac{\left(4 \times \frac{1}{4}\right) \pi r^2}{a^2}$

packing effeciency/packing fraction PF = π/4

$= \frac{\pi r^2}{4r^2} \times 100$

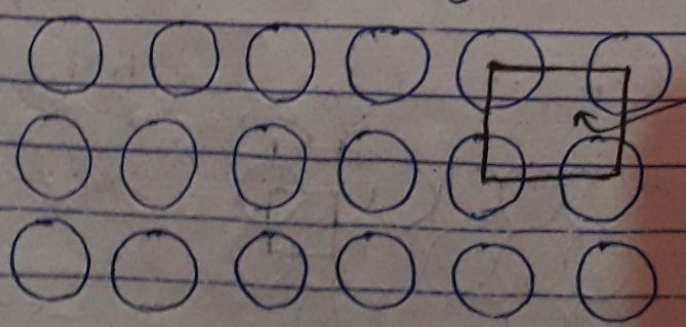
$= 25\pi$

$\approx 78\%$

Important point :-

When such 2D packing with touching spheres is extended in 3D, it form simple cubic structure (or primitive cubic structure)

ii) With non-touching spheres :-



* Coordination Number = 4

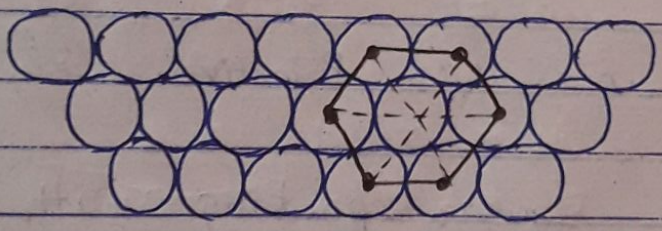
* Packing efficiency can be calculated if relation between a & r is known.

efficiency will be less than square packing with touching spheres

Important point :-
Square packing with non - touching spheres when extended in 3D, forms body centred cubic structure.

packing effeciency/packing fraction PF = π/4

b. Hexagonal Packing.



* Coordination number = 6

* Area efficiency = $\frac{(6 \times 1/3 + 1)\pi r^2}{6 \times \sqrt{3}/4\, a^2} \times 100$

$= \frac{50 \times \pi}{\sqrt{3}}$

$= \frac{50 \times 3.14}{1.73}$

$\approx 91\%$

packing efficiency/packing fraction PF = $\pi/2\sqrt{3}$

Important point :-

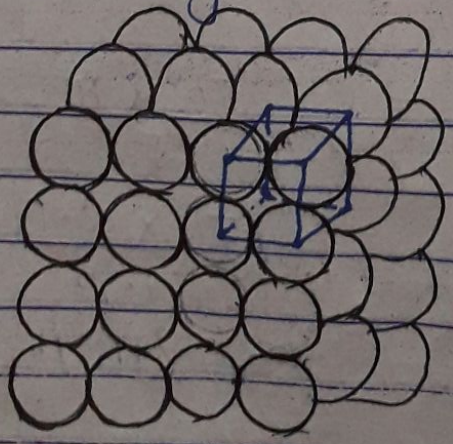
- Hexagonal 2D packing is more efficient than the square 2D packing.
- This hexagonal 2D packing when extended in 3D, forms hcp or ccp structure depending upon stacking pattern.

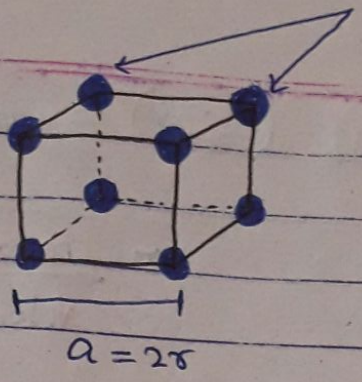
(*4) * Packing of identical sphere in 3D {metallic crystal}

@ Simple Cubic {or Primitive Cubic} Structure :-

→ each layer is square packing with touching sphere.

→ 2nd, 3rd and subsequent layers are similar layers placed eclipsing to the first layer (AAAAA.....)





Spheres in simple cubic touch along the edge.

r = radius of sphere
a = edge length of cubic unit cell.

Simple Cubic unit cell
(or Primitive Cubic unit cell).

* Coordination number = 6 { each sphere touches 4 in its own plane, 1 in higher & 1 in lower plane }

* Effective no. of spheres in simple cubic unit cell.

$= \frac{1}{8} \times 8$

$= 1$

* Packing efficiency $=$ $\frac{\text{Volume of spheres inside unit cell}}{\text{vol. of unit cell}} \times 100$

$= \frac{1 \times \frac{4}{3}\pi r^3}{a^3} \times 100$

packing efficiency/packing fraction PF = π/6

$\approx$ 52.3 %.

⊛ Density of Crystal (sc) = density of unit cell

$$= \frac{\text{mass of unit cell}}{\text{volume of unit cell}}$$

mass of 1 atom of an element in grams

$$= \frac{(1 \times M/N_A)\ \text{gram}}{a^3\ (cm^3)} \quad g/cc.$$

{M = molar mass}

mass of 1 atom of an element in amu

$$= \frac{(1 \times M)\ amu}{a^3\ (A^{\circ 3})} \quad , amu/A^{\circ 3}$$

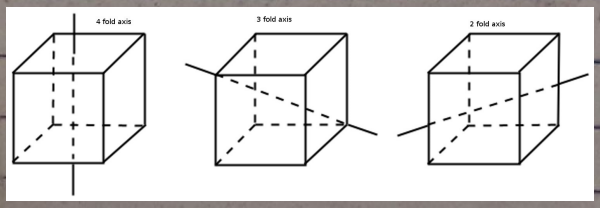
⊛ for metallic crystal,

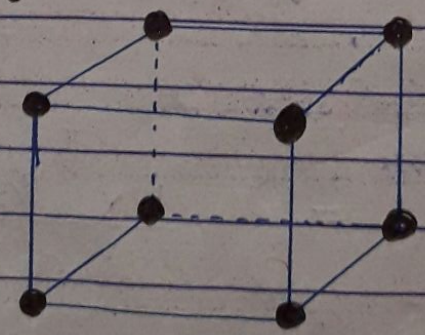
$$d = \frac{Z \times M/N_A}{a^3} \quad , g/cc$$

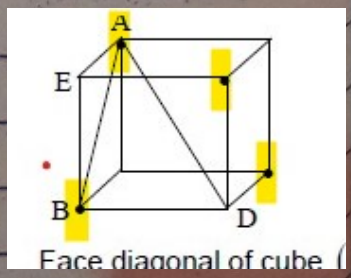
where, M = molar mass, g/mol
Z = effective no. of atoms in unit cell.

(*5)

→ Properties of Cube :-







→ 8 Corners ;- each Corner is equally shared in 8 Cubes.
→ 1/8 th portion inside.

→ 6 Square faces :- each face is shared in 2 Cubes (so each face centre & face diagonal is shared in 2 cubes).

→ 1/2 portion inside

→ 12 edges ~~each~~ :- each shared in 4 Cubes.

→ 1/4 th portion inside.

→ 4 body diagonals :- Unshared.

Q: In a Cubic crystal structure, atoms of X occupy all corners, atoms of Y occupy all face centre & atoms of Z occupy body centre. Find :-

a: formula of Compound
b: density in g/ml & in amu/$A^{\circ 3}$.

Ans :- Given, Edge length, $a = 2A^{\circ}$

$X = 20$
$Y = 40$
$Z = 60$
} Amu

$N_A = 6 \times 10^{23}$.

a: $X_{\frac{1}{8} \times 8}\ Y_{\frac{1}{2} \times 6}\ Z_1 = XY_3Z$ Ans.

formula of Compound.

b. $d\,(\text{in g/ml}) = \dfrac{\dfrac{20 + 40\times3 + 60}{6\times10^{23}}\ g}{(2\times10^{-8})^3\ cm^3}$

$= \dfrac{200}{6\times10^{23}\times8\times10^{-24}}\ g/cc$

$= \dfrac{200}{6\times8\times0.1}\ g/cc$ (100 → 25)

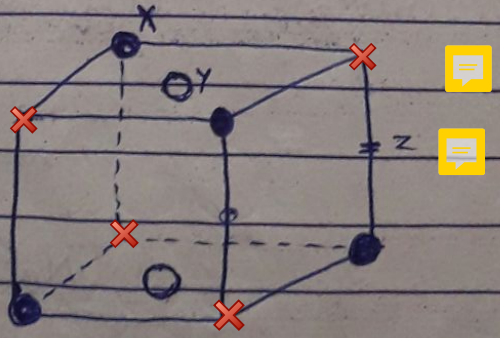
$= \dfrac{125}{3}\ g/cc.$

& $d\,(\text{in Amu/A}^\circ) = \dfrac{200}{2^3}\ \dfrac{amu}{A^{\circ 3}}$

$= 25\ amu/A^{\circ 3}$

Q. In above question X occupy alternate Corners, Y occupy all alternate face Centres & Z occupy all edge Centres. Solve the question.

Ans.



a. $X_{1/8\times4}\ Y_{\frac{1}{2}\times2}\ Z_{\frac{1}{4}\times12} = X_{\frac{1}{2}}\ Y_1\ Z_3$

$= X_1 Y_2 Z_6$ Ans

b. $d\ (\text{in } g/cc) = \frac{\frac{1}{2}\times 20 + 1\times 40 + \frac{3}{4}\times 60 \text{ gfc}}{6\times 10^{23}\times (2\times 10^{-8})^3}$

$= \frac{230}{6\times 8\times 0.1}\ g/cc$

$= \frac{2300}{48}\ g/cc$

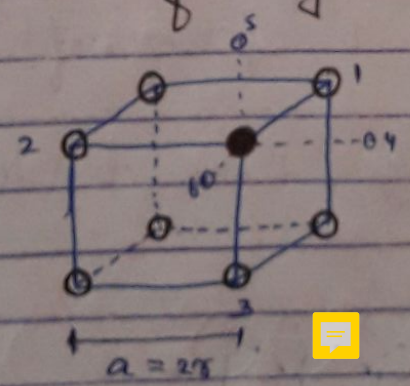
$= 47.91\ g/cc.$

& $d\ (\text{in } Amu/A^\circ) = \frac{230\ g\ amu}{2^3\ A^{\circ 3}}$

$= \frac{230}{8}\ g\ amu/A^\circ$

$= 28.75\ amu/A^\circ$

(#6) * Calculation of neighbours in simple cubic struct.



No of nearest neighbours (C.N) = 6 at 'a'

Alternate Method :-

• No. of nearest neighbours (C.N) $= \frac{3}{4}\times 8$

$= 6$ at a dist. 'a'

- No. of next neighbours. **or 2nd coordination no.**

$= \frac{3}{2} \times 8$

$= 12$ at '$\sqrt{2}a$'

- no. of next to next nearest neighours **or 3rd coordination no.**

$= \frac{1}{1} \times 8$

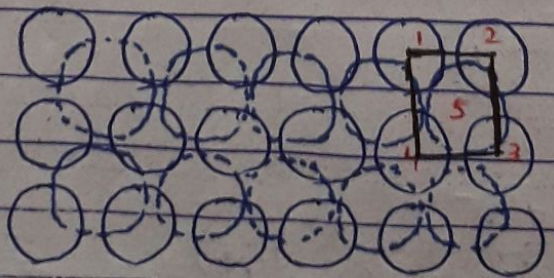
$= 8$ at distance '$\sqrt{3}\,a$'.

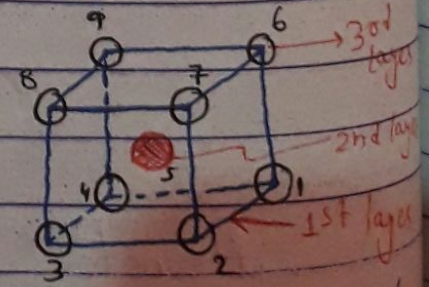
(b) Body Centred Cubic structure : (bcc)

→ Each layer is square packing with non-touching spheres.

→ Spheres of second layer is placed into the depression of 4 spheres of first layer.

→ Third layer exactly same to the first layer.





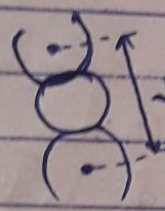
spheres in bcc touch along body diagonal.

⊛ Coordination number = 8 { each sphere touches 0 in down, 4 in higher & 4 in lower }

⊛ Effective no. of sphere in bcc unit cell

$$Z = \frac{1}{8} \times 8 + 1 = \underline{2}$$

⊛ Packing efficiency = $\frac{2 \times \frac{4}{3}\pi r^3}{a^3} \times 100$



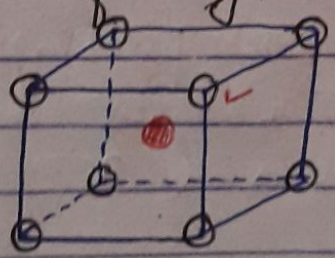
= 68%

{ Better packing then simple Cubic structure }

⊛ Density of bcc lattice

$$= \frac{(2 \times M/N_A)}{a^3} \text{ g/cc.} \quad \{Z = 2\}$$

* Calculation of neighbours in bcc structure :-



(bcc unit cell)

→ No. of nearest neighbours = C.N = $\frac{1}{1} \times 8$ = 8 at $\frac{a\sqrt{3}}{2}$

→ No. of next nearest neighbours =

$= \frac{3}{4} \times 8$

= 6 at distance 'a'.

→ No. of next of to next nearest neighbours :-

$= \frac{3}{2} \times 8$

= 12 at distance '$a\sqrt{2}$'

(*7) Closed Packed Structure :
{ Hexagonal closed packed (hcp) & Cubic closed packed (ccp) } :-

→ Both hcp & ccp structure are formed using hexagonal 2D layers. Every layer in hcp and ccp are hexagonal 2D layer.

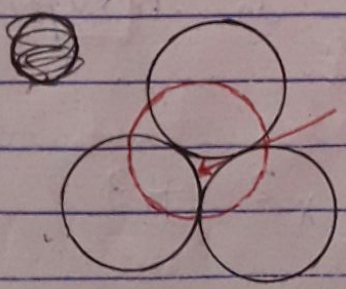
→ hcp & ccp are called closed packed structure because packing efficiency sub stacking pattern is max with identical spheres ($\approx 74\%$)

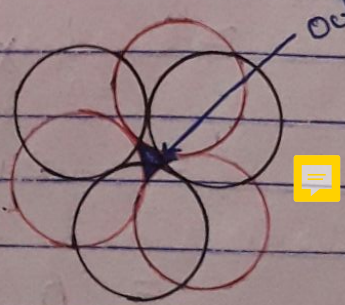
→ Spheres of 2nd layers are placed into the depression of 1st layer (hexagonal 2D layer)

This forms or generates two types of Voids, known as tetrahedral voids (TV_s) & octahedral voids (OV_s).

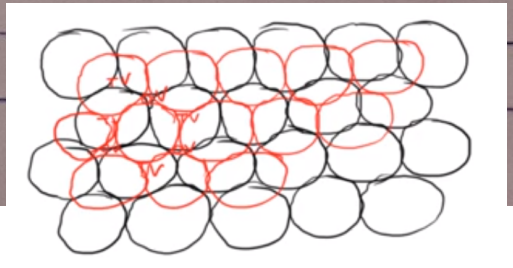
→ If spheres of 3rd layers are placed on TV_s then 3rd layer become similar (or eclipsing) to the first layer and stacking arrangement becomes ABABAB..... (This generates hcp structure).

→ If spheres of 3rd layers are placed on OV_s, then 3rd layer is different from 1st & 2nd layer & stacking arrangement becomes ABC ABC ABC..... (This generates ccp also known as face centred cubic, fcc structure)

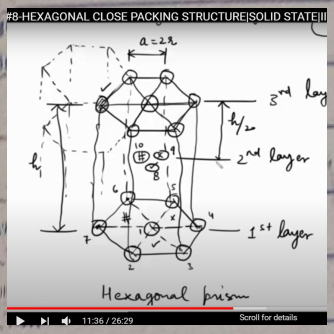


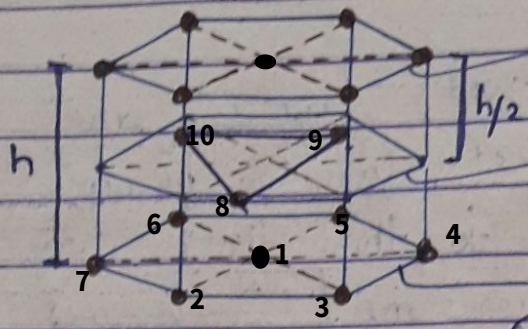


Around every spheres 3 TV_s & 3 OV_s are present



(*8) Hexagonal closed packed structure :- (HCP)
(ABABABAB.....) {hexagonal 2D layers}





hexagonal prism (HCP unit cell)

(*) Coordination number = 12 {each sphere touches 6 in own plane, 3 in higher & 3 in lower plane}

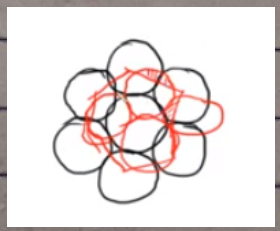
with identical spheres, the maximum possible CN is 12
the CN with identical spheres cannot be more than 12

(*) Effective no. of spheres in hexagonal prism (or hcp) unit cell.

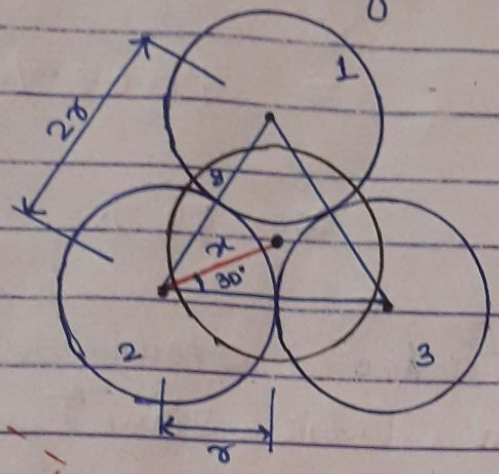
$= \frac{1}{6} \times 12 + \frac{1}{2} \times 2 + 3$; taken Completely inside, however their some portion is out of hexagonal prism

$= 2 + 1 + 3$

$= 6$

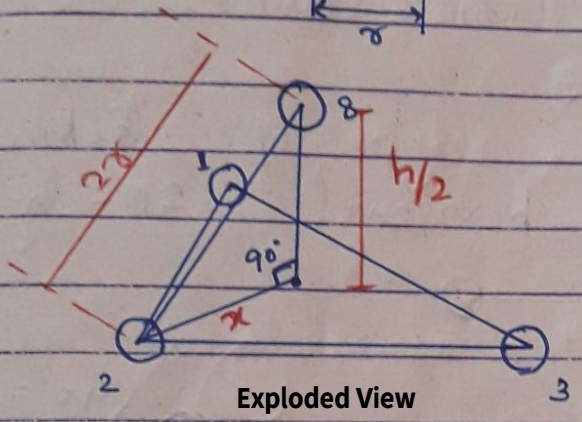


⊛ Calculation of 'h' :-



$$\cos 30^\circ = \frac{r}{x}$$

$$x = \frac{2r}{\sqrt{3}}$$



Exploded View

$$\frac{h}{2} = \sqrt{(2r)^2 - x^2}$$

$$= \sqrt{(2r)^2 - \left(\frac{2r}{\sqrt{3}}\right)^2}$$

$$\boxed{h = 4r \times \sqrt{\frac{2}{3}}}$$

$$\Rightarrow h < 4r \quad \{\text{I.I.T}\}$$

{ It is because spheres are placed into the depression of first layer & not kept one over another }

⊛ Packing efficiency $= \dfrac{6 \times \frac{4}{3}\pi r^3}{6 \times \frac{\sqrt{3}}{4} \times (2r)^2 \times h}$ ——— (1)

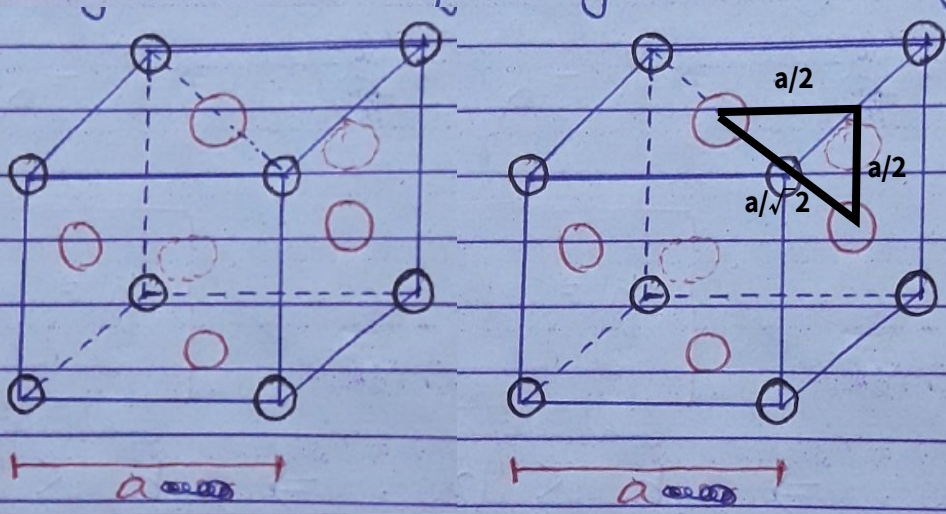
from eq (1), packing efficiency

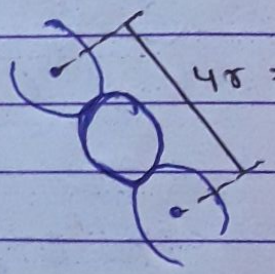
$= 74.05\ \%$

$\approx 74\ \%$. (= max. possible efficiency with identical spheres)

(*9) Cubic Closed Packed Structure (CCP):
{ face centred cubic structure (FCC) }

→ ABC ABC ABC..... type of stacking with every layer as hexagonal 2D-layer.





$$a = \frac{4r}{\sqrt{2}}$$

FCC unit cell
{ In FCC structure spheres touch along face diagonal }.

* Coordination number = 12 { each sphere touches 6 in own plane, 3 in higher & 3 in lower plane }

or, alternatively,
each sphere touches 4 in own, 4 in lower and 4 in higher plane

in FCC, every sphere at $a/\sqrt{2}$ distance must touch

⊛ Effective no of atoms in fcc unit cell

$$= \frac{1}{8} \times 8 + \frac{1}{2} \times 6$$

$$= 4$$

⊛ Packing efficiency $= \frac{4 \times \frac{4}{3}\pi r^3}{a^3} \times 100$

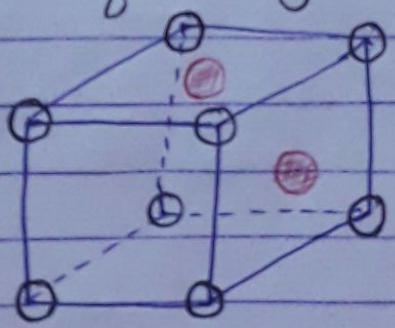
$$= \frac{4 \times \frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3} \times 100$$

$$\approx 74.05\ \%$$

$$\approx 74\ \%$$

⊛ density $= \frac{4 \times (M/N_A)}{a^3}$, g/cc

* Calculations of neighbours in FCC :-



FCC

→ No. of nearest neighbours $(C_N) = \frac{3}{2} \times 8$

$= 12$ at $(a/\sqrt{2})$

→ no. of next nearest neighbours =

$$= \frac{3}{4} \times 8$$

= 6 at distance 'a'

→ no. of next to next nearest neighbour
{ remaining 3 face centre sphere } =

$$= \frac{3}{1} \times 8$$

= 24 at distance '$a\sqrt{1.5}$'

(*10) Visualising Various planes in Fcc :-
{ ccp ≅ Fcc }

(11) Numerical problems on Metallic Crystals

Note:-

→ SC {simple cubic} : eg → Po (Polonium) (rare)

→ BCC : eg → Li, Na, K, Rb, Cs etc

→ HCP : eg → Mg, Zn etc

→ Fcc : eg:→ Cu, Ag, Au etc (most Common)

Q. Aluminium (27) crystallizes in cubic system and edge length of cube unit cell is 4A°. Density of Al = 45/16 g/cc, find atomic radius of Al atom.

($N_A = 6 \times 10^{23}$)

Sol^n

$$d = \frac{45}{16} = \frac{Z \times 27/6\times10^{23}}{(4\times10^{-8})^3}$$

$$\frac{45}{16} = Z \times \frac{9}{2\times64\times0.1}$$

$$\boxed{Z = 4} \rightarrow \text{FCC}$$

$4r = a\times\sqrt{2}$ {Property of Fcc}

$r = \frac{4\sqrt{2}}{4} = \sqrt{2}\,A^\circ$ Ans

Q. A metal crystallizes in cubic system (edge length of unit cell = 4Å) density is 6.25 g/cc. Find no. of unit cell in 200 g of matter.

Soln $a = 4Å$, $d = 6.25$ g/cc

$$\Rightarrow 6.25 = \frac{Z \times M/6\times10^{23}}{64\times10^{-24}}$$

$$\Rightarrow Z \times M = 6.25 \times 6.4 \times 6$$

$\because$ M g metal $\longrightarrow \left(\frac{6\times10^{23}}{Z}\right)$ no. of unit cells

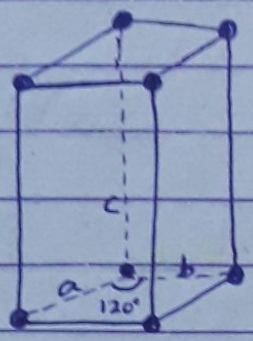
$\because$ 200 g metal $\longrightarrow \frac{6\times10^{23}}{Z\times M} \times 200$

$$\longrightarrow \frac{6\times10^{23} \times 200}{6.25 \times 6 \times 6.4}$$

$\longrightarrow 5\times10^{23}$ unit cells

Ans

Q. Ice crystallizes in hexagonal lattice. At low temp. density of ice is 0.92 g/cc. Lattic Constant are $a = b = 4.53\ A^\circ$, $c = 7.41\ A^\circ$ Find no. of H_2O molecules Contained in one unit cell of ice.



Soln

$$d = 0.92 = \frac{Z \times 18}{6 \times 10^{23} \times V_{unit\ cell}}$$

$$\Rightarrow 0.92 = \frac{Z \times 18}{6 \times 10^{23} \times \left(2 \times \frac{\sqrt{3}}{4} a^2\right) \times c}$$

$$\Rightarrow 0.92 = \frac{Z \times 18}{6 \times 10^{23} \times \frac{\sqrt{3}}{2} \times (4.53 \times 10^{-8})^2 \times 7.41 \times 10^{-8}}$$

$$\Rightarrow \boxed{Z = 4} \quad \text{Ans}$$

(*12) → IONIC CRYSTALS : { eg → Ionic Compounds like NaCl, KCl, CuS etc }

* some covalent compounds (ex. diamond in ZnS structure) crystallize like ionic compounds

→ In ionic Compounds we have to learn the packing of two different sized ions (Cation & anions).

→ Anions being usually larger in size form lattice & Cation (or smaller ion) occupies void space or intustitial space.

→ The type of void occupied by smaller ion depends on the relative size of Cation & anion.

r_+ → radius of cation
r_- → radius of anion

$0.155 = \frac{r_+}{r_-}$ → radius ratio (RR) = 0.155

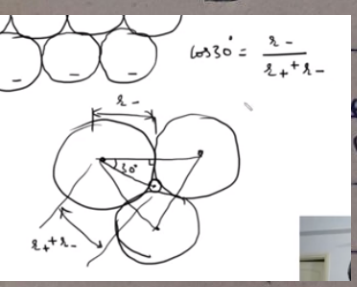
↳ critical or minimum radius ratio { correspond to ideal or undistorted crystal }

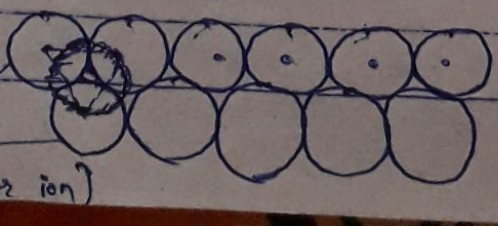
* Types of Voids :-

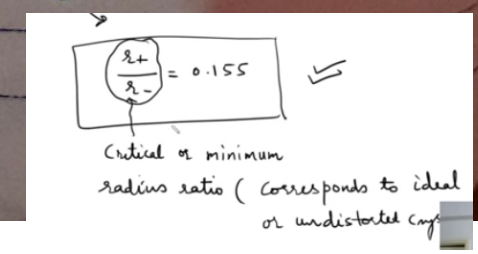
0. Linear void

(1) Trigonal planar void (CN = 3)

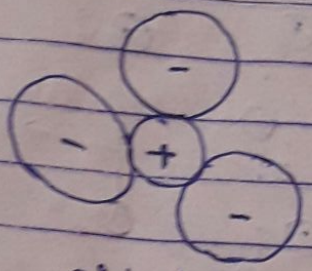
eg:- Hexagonal 2D layer







Actual (or distorted crystals): $C_N = 3$



Distorted crystal

→ Radius Ratio

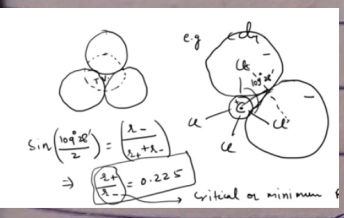
$\frac{r_+}{r_-} > 0.155$

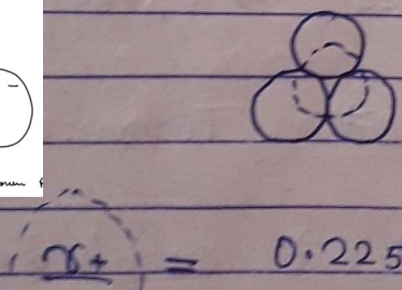
eg:- 0.170

→ Ideal crystal is infact not desirable as P.E will be high (& hence stability will be low) if anions also touch in required geometry.

(2.) Tetrahedral void ($C_N = 4$)

{ 3D void }





ex. CCl4

C acts like void

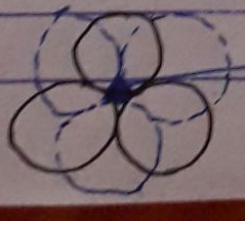
Cl acts like lattice ions

$\frac{r_+}{r_-} = 0.225$

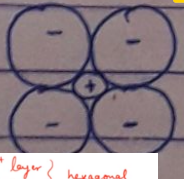
→ Critical or minimum radius ratio

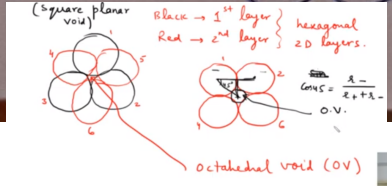
(*13) (3) Octahedral void (in 3D) ($C_N = 6$)

Square planar void (in 2D)



void



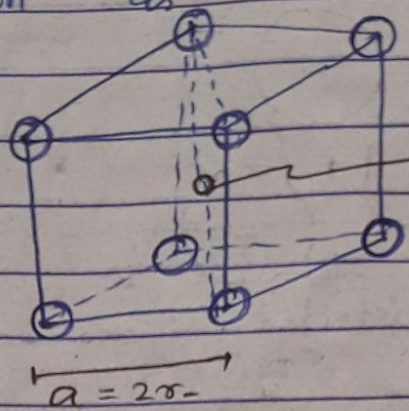


$\frac{r_+}{r_-} = 0.414$

critical or minimum radius ratio

(4) Cubical Void (CN = 8)
(also known as Bcc void)

$$(2r_+ + 2r_-)^2 = (\sqrt{2} \times 2r_-)^2 + (2r_-)^2$$



$$\frac{r_+}{r_-} = 0.732$$

Summary :-

Radius Ratio — Void occupied

$0.155 \leq R.R < 0.225 \Rightarrow$ Trigonal planar void

$0.225 \leq R.R < 0.414 \Rightarrow$ Tetrahedral void

$0.414 \leq R.R < 0.752 \Rightarrow$ octahedral void

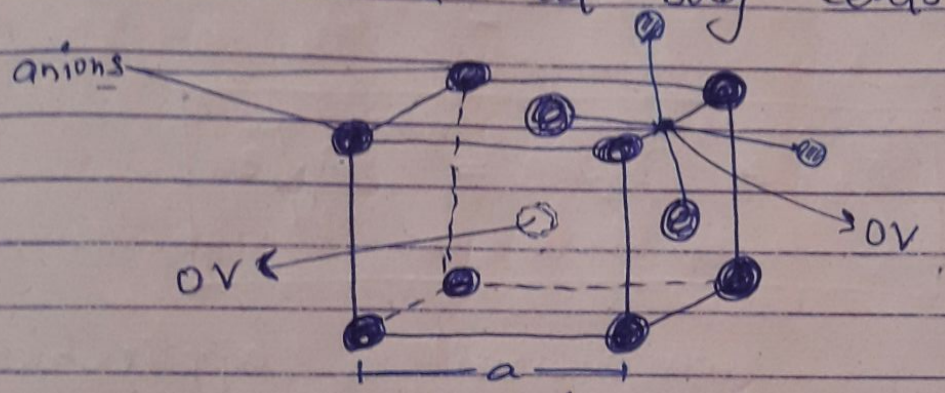
$0.732 \leq R.R \Rightarrow$ Cubical void

(#14) * Tetrahedral & octahedral voids in Close packed structure :-
{ HCP & FCC }

(a) In FCC

i) Octahedral voids :- (CN = 6)

→ OV in FCC occurs at all the edge centre & at body centre.



FCC unit cell

VVI

$$r_+ + r_- = a/2$$

Compulsorily followed.

→ Effective no. of octahedral void in FCC unit cell = $\frac{1}{4} \times 12 + 1$

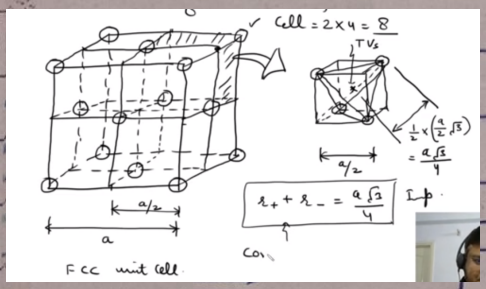
$= 4$

→ Minimum distance between two octahedral voids in FCC = $a/\sqrt{2}$.

ii) Tetrahedral Voids in Fcc :-

→ TVs occurs in Fcc along body diagonal (there are two TVs on each body diagonal).

∴ total no. of TVs in one Fcc unit cell = 2 × 4
= 8



$$r_+ + r_- = \frac{a\sqrt{3}}{4}$$

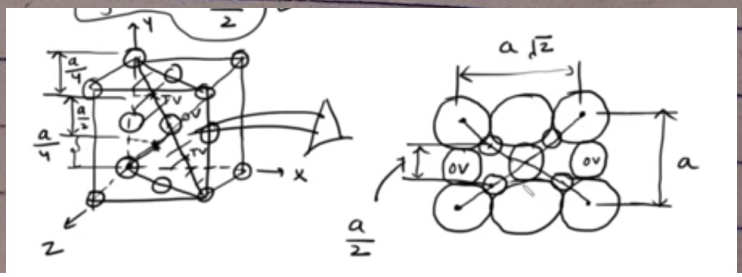
Compulsorily applicable for T.Vs in Fcc.

(edge length = a)

→ If a larger cube is divided into smaller cube of edge length a/x then no of cubes from.

$= x^3$

→ Minimum distance between two TVs in Fcc = $a/2$



the TV and OV do not touch each other

(*15) (b.) In HCP :-

* Octahedral & Tetrahedral voids.

In General in closed packed structure (HCP & FCC)

No. of OV = N
& No. of TV = 2N.

{ where N = effective no. of atoms in one unit cell.

In, FCC N = 4
⇒ No. of OV = 4
& No. of TV = 8

Similarly In HCP N = 6
⇒ no. of OV in HCP = 6
& no. of TV in HCP = 12.

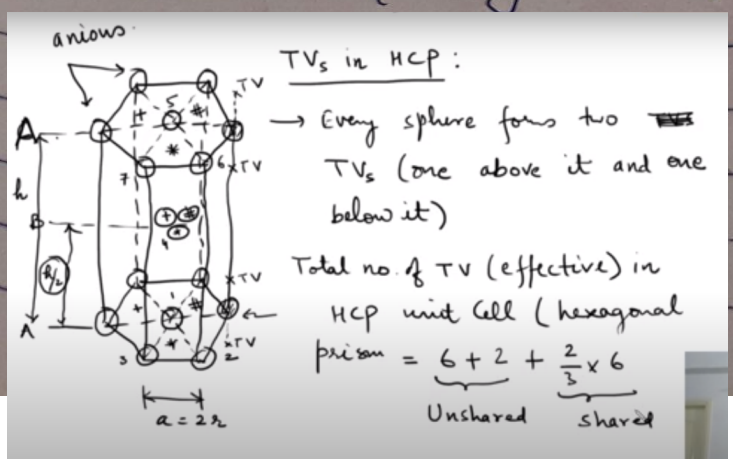
* TVs in HCP :-

→ Every sphere forms two TVs (one above it & one below it).

Total no of TV (effective) in HCP unit cell {hexagonal prism}

$= \underbrace{6}_{\text{unshared}} + 2 + \underbrace{\frac{2}{3} \times 6}_{\text{shared}}$

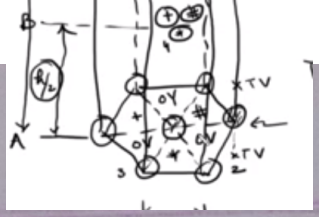
$= 12$

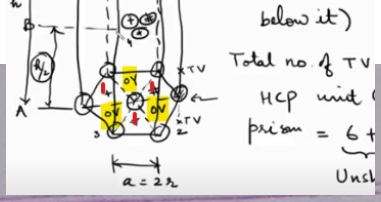


TVs in HCP :

→ Every sphere forms two TVs (one above it and one below it)

Total no of TV (effective) in HCP unit Cell (hexagonal prism = $\underbrace{6}_{\text{Unshared}} + 2 + \underbrace{\frac{2}{3} \times 6}_{\text{shared}}$

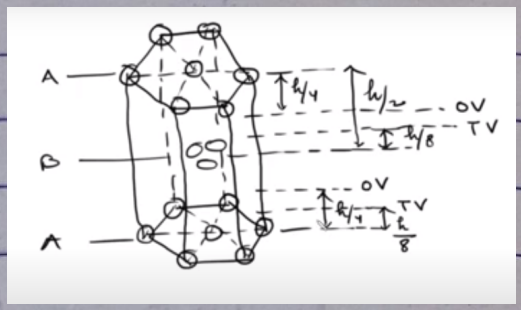




Total no. of OVs in HCP unit cell :-

$= \frac{6}{2}$

Unshared.



→ TVs are found at h/8 height from every layer.

→ OVs are found at h/4 height from every layer.

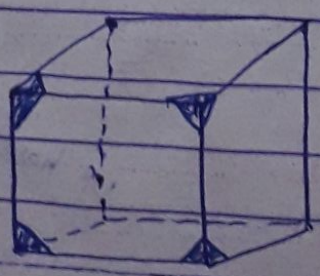
→ where h = Centre to centre distance between 1st & 3rd layer.

or height of HCP.

(*16) * Truncated Polyhedron :-

↳ means Cutting the vertices (or corner) of a polyhedron.

eg:- * Truncated Cube :-



Square face is converted into Octagon

↳ 8 sides.

In Truncated Cube

⇒ 6 → octagonal faces

⇒ 8 → triangular faces

14 → total faces.

* Truncated tetrahedron :-

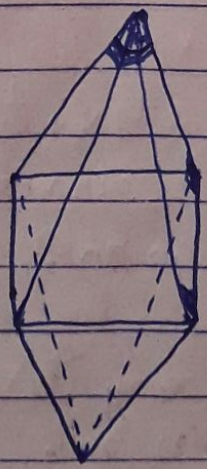
↳ no. of hexagonal faces = 4

↳ no. of triangular faces = 4

total = 8

IIT Q. Find no. of hexagonal faces present in a truncated octahedron.

Sol^n



Ans :- 8

no. of hexagonal faces = 8

" " square faces = 6

total 14.

(*17) * Various important Ionic Crystal :-

(i) Rock salt (NaCl) structure :-

$\frac{r_{Na^+}}{r_{Cl^-}} = 0.52$

{ octahedral void will be occupied }

→ Anions form ccp lattice & Cation occupy all the octahedral voids.

→ There are 4 cation & 4 anion per unit cell so simple formula is AB type.

→ $r_+ + r_- = a/2$

↳ always applicable

→ Coordination number of Cation = 6

" " " anion = 6

Ratio of C.N of Cation & anion is equal to the inverse ratio of Cation & anion. moles

eg :- for AB Crystal (1:1) molar ratio ⇒ $\frac{CN \text{ of } A^+}{CN \text{ of } B^-} = \frac{1}{1}$

• AB_2 Crystal (1:2) type ⇒ $\frac{C.N \text{ of } A^{2+}}{CN \text{ of } B^-} = \frac{2}{1}$

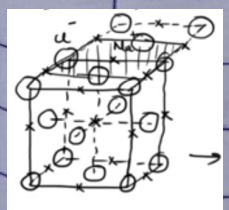
→ Packing efficiency = $\frac{4 \times \frac{4}{3}\pi r_+^3 + 4 \times \frac{4}{3}\pi r_-^3}{a^3} \times 100$

→ Density of crystal = $\frac{4 \times \frac{M_{AB}}{N_A}}{a^3}$, g/cc

→ Other examples include :-
- Halides of Alkali metals except halides of Cs.
- Oxides of alkaline earth metals except 'BeO'.

→ In NaCl, Cl^- form ccp & Na^+ occupy octahedral voids.

{ Alternatively, Na^+ form ccp & Cl^- occupy octahedral voids.



or

Cl^- form ccp, Na^+ form ccp & both are inter penetrated to form rock salt structure.

→ for AB types of structure.

ZnS → Zinc blende (ccp) ✓
→ Wurtzite (HCP) ✗

(*J8) (ii) Zinc blende (ZnS) structure:

→ $\frac{r_{Z^{2+}}}{r_{S^{2-}}} = 0.4$ {tetrahedral void}

→ S^{2-} forms ccp & Z^{2+} occupies alternate tetrahedral void.

→ There are four cation & 4 anion per unit cell so Simplified formula is AB types.

→ $r_+ + r_- = \frac{a\sqrt{3}}{4}$

↳ always applicable.

→ C.N of cation = 4
" " anion = 4

→ Density 'd' = $\frac{4 \times \frac{M_{AB}}{N_A}}{a^3}$ g/cc

→ Packing efficiency = $\frac{4 \times \frac{4}{3}\pi r_+^3 + 4 \times \frac{4}{3}\pi r_-^3}{a^3} \times 100$

→ Other example include
• diamond, $r+r = \frac{a\sqrt{3}}{4}$
↳ No. of 'C' atom in one unit cell = 8

- HgS $\left\{ r_+ + r_- = \frac{a\sqrt{3}}{4} \right\}$

(iii) Cesium chloride structure ($CsCl$):-

(AB)

→ $\frac{r_{Cs^+}}{r_{Cl^-}} = 0.93$ {Cubical void is occupied}

→ Anions occupy all 8 Corners of cube & ~~Centre~~ Cation occupies body Centre.

→ There is one cation & one anion per unit cell, so formula is AB type.

→ $r_+ + r_- = \frac{a\sqrt{3}}{2}$

→ Density 'd' = $\frac{1 \times M_{AB}/N_A}{a^3}$, g/cc

→ Other examples include,
- $CsCl$, $CsBr$, CsI

imp.

→ Under high pressure condition $RbCl$, $RbBr$, RbI also becomes 8:8 (ie Caesium chloride) structure.

Memorise: (Imp), P↑, C.N↑ (e.g 6:6 → 8:8); T↑, C.N↓ (e.g 6:6 → 4:4)

→ On increasing Pressure, C.N Increase & on increasing Temp, C.N decrease.

Flourite

(iv) (CaF_2) structure :-

→ Ca^{2+} (or Cation) forms ccp lattice & F^- (or anion) occupy all TVs.

→ There are 4 cation & 8 anion per unit cell so formula is AB_2 type.

→ $r_+ + r_- = \frac{a\sqrt{3}}{4}$ always applicable

→ Density 'd' $= \frac{4 \times \frac{M_{AB_2}}{N_A}}{a^3}$, g/cc.

→ C.N of F^- = 4
& C.N of Ca^{2+} = 8

→ Other examples include
• SrF_2, $BaCl_2$, BaF_2, $BaCl_2$ etc

(v) Antifluorite structure (Na_2O) :-

→ Anions form ccp lattice & Cations occupy all TVs.

→ There are 8 Cation + 4 anion } in 1 unit cell
{ A_2B type }

→ $r_+ + r_- = \frac{a\sqrt{3}}{4}$

→ other example include
↳ Li_2O, Na_2O, K_2O etc.

Some mixed oxide structure.

a. The spinal structure :-
{ natural mineral, spinel ; $MgAl_2O_4$
or $MgO \cdot Al_2O_3$ }

O^{2-} ⟶ ccp
Mg^{2+} ⟶ $\frac{1}{8}$ th of TVs.
Al^{3+} ⟶ half of OVs.

} $Mg.Al_2O_4$

b. The perovskite :- ($CaTiO_3$) str :-

(a mineral.

O^{2-} & Ca^{2+} form ccp

$CaTiO_3$

Ti^{4+} → occupies all corners of cube

O^{2-} → all edge centres of cube.

Ca^{2+} → body centre of cube

CaO.TiO2
mixed oxide

* Defects (or imperfection) in crystalline solids :-

→ At absolute zero (i.e zero kelvin), there are no defects and when temp. increases, defects in crystalline solids increases. so, defects are known as "thermodynamics defects".

→ Defects in solids may be of many types like point defects, line defect, surface defect & volume defects.

* we have to study only point imperfection as these are important in ionic crystals.

* other types of defects (line defects, surface defects etc) are found in metallic crystals.

* Point defects in Ionic solids :-

↳ There is an imperfection at a lattice site in ionic solid.

a. stoichiometric defects → Shottky defect
→ Frenkel defects

b. Non- Stoichmetric defects

- Metal excess defects
- Metal deficiency defects

c. Impurity defects

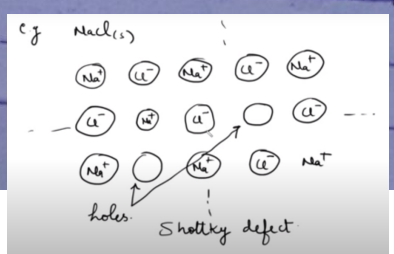
- Substitutional impurity defects
- Interstitial impurity defects

a. Stoichiometric defects :- formula or stoichiometry remains unchanged due to such defects.

i) **Shottky defect :-** In shottky defects, Cation & anions are found to be missing at certain lattice site in appropriate numbers, so that crystal is electrically neutral forming holes.

eg:- • NaCl, equal no of Na^+ & Cl^- are missing in NaCl crystal at room temp.

• $CaCl_2$, for each cation (Ca^{2+}) two anions ($2Cl^-$) are missing.



→ Infact defects are desirable in solids to improve their electrical & mechanical properties (eg:- ductility / malleability / strength / hardness etc).

eg:- • To increase hardness and strength of Fe metal, a certain amount of Carbon should be present in ~~inst~~ interstitial position of Fe lattic (interstitial defect).

• Si, Ge (intrinsic semiconductor) to improve electrical Conductance, intentionally a dopping of B, Ga, As etc is done.

→ Electrical Conductance of such solids is due propagation of holes under electric field (movement of hole under electric field is equivalent to movement of ions in opposite direction).

→ Shottky defat decreases density of solids.

$$\% \text{ of vacancies in crystal} = \frac{d_{ideal} - d_{observed}}{d_{ideal}} \times 100$$

$d_{observed} < d_{ideal}$

} ideal density corresponds to 100% occupancy of ions @ lattice sites

→ This defects arises when size of cation & anion are nearly same (or size difference is small).

eg:- NaCl , KCl , CsCl , $CaCl_2$, AgBr

(6:6) (8:8) (2:4)

Shows Shottky defect as well as Frankel defect.

$\Delta S = k \ln p$

where ΔS = Entropy change due to shottky defect

(*20)

ii) **Frenkel defects :-**

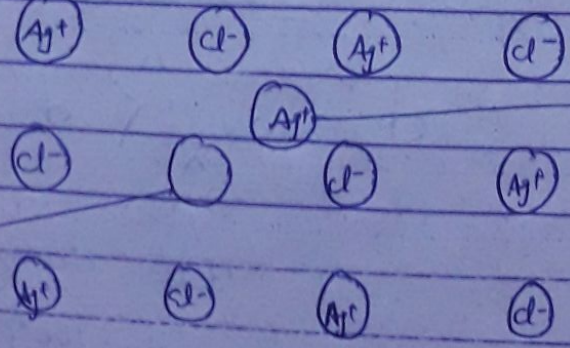
→ This arise when size difference between cation & anion is large.

→ In this defect, Cation (being smaller in size) leaves its lattice point (site) and occupies the nearby ~~inst~~ intustitial site.

4:4 4:4 4:4 4:4 (C.N)

eg:- AgCl , AgBr , AgI , ZnS

shows both Schottky and Frenkel



Frankel defect

→ Frankel defect :- A Combination of Vacancy (hole) & intustitial defect.

* Intustices means Voids.

→ density remains Unaffected.

→ Due to accumulation of similar charge (locally), polarity induces in the crystal and dielectric Constant is increased.

→ Lattice energy ↓ due to both Frenkel & Shottky defect.

→ Electrical Conductance increases. due to the creation of holes

b. Non - stoichiometric defects :-

i) **Metal excess defect :-**

In such defect, an electron occupies the lattice site of anion insted of anion.

eg:- $NaCl \rightarrow Na \xrightarrow{\Delta}$ (vapour)

$Na \longrightarrow Na^+ + (e^-)$

$Na^+ + Cl^- \longrightarrow NaCl$

↳ This electron diffuses in the crystal & occupies the Cl^- lattice size form F - Centres.

F - Centre → Farbencentre
or Coloured Centre.

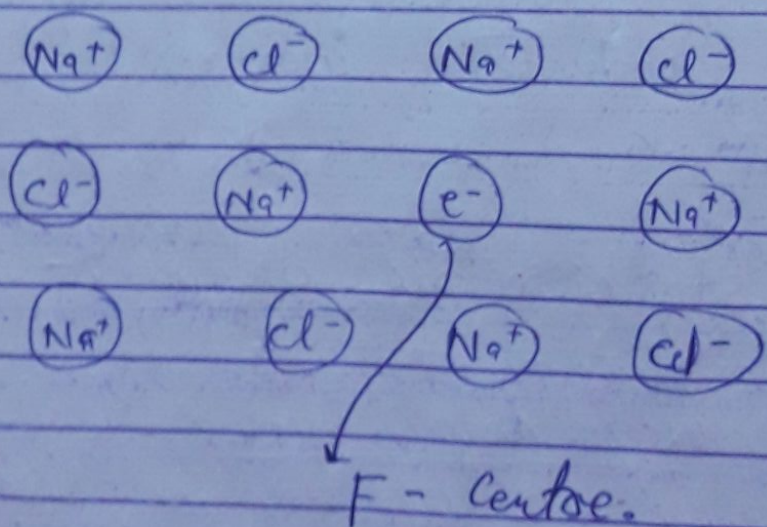
→ Solids become paramagnetic & Coloured & Conductive.

eg:- • NaCl $\xrightarrow[\Delta]{Na(v)}$ Yellow
{white when cold}

• LiCl $\xrightarrow{\Delta}$ Pink

• KCl $\xrightarrow{\Delta}$ Lilac or violet

• ZnO $\xrightarrow{\Delta}$ Yellow
{white when cold} {when hot}
loses oxygen on heating



F - Centre.

ii) **Metal deficiency defects :-**

→ Many ionic Compounds (ores / minerals) are found in earth Crust in non-stoichiometric proportion.

eg:- • FeO is found in earth Crust as $Fe_{0.95}O$ to $Fe_{0.97}O$

FeO ⇝ Stoichiometric Compound
1:1

$Fe_{0.97}O$ ⇝ Non-Stoichiometric Comp.

Note :- Every Compound, be it stoichiometric or non-stoichiometric must be electrically neutral.

Total +ve charge = | Total -ve charge |

→ This ensures that in $Fe_{0.97}O$ or $Cu_{1.7}S$ iron is present as Fe^{2+} & Fe^{3+}
Also in $Cu_{1.7}S$, Cu is present as Cu^{+} & Cu^{2+}

Q. Find % of Fe^{2+} of total iron in $Fe_{0.97}O$

Ans $Fe_{0.97}O \rightarrow Fe^{2+} + Fe^{3+} = 0.97$ mole

$Fe_{0.97}O$: 0.97 mole; O: 1 mole; Fe^{2+}: (x mole); Fe^{3+}: (0.97 − x) mole

⇒ $2x + 2.91 - 3x = 2$
$-x = 0.91$ mole.

% of Fe^{2+} of total iron
$= \frac{.91}{.97} \times 100$

(21)

c. Impurity defects :- {Non - stoichiometric defect}

i) Substitutional impurity defect :

→ This arise when doping of smiar sized cation is done to substitute the existing cation of ionic crystal.

eg:- • when molten NaCl is crystallised having a small amount of $SrCl_2$, then each Sr^{2+} displaces two Na^+ ions (size of Sr^{2+} is nearly same as that of Na^+ ion). one Na^+ position is occupied by Sr^{2+} & other is left as vacency.

In this case, No. of vacancies generated = no. of Sr^{2+} doped in NaCl.

• when AgCl is doped with $CdCl_2$, No. of vacancies generated = no. of Cd^{2+} ions doped.

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eg:- when Li_2O is doped in $Ni^{+2}O^{2-}$ crystals then one Li^+ replace one Ni^{2+} but to maintain electrical neutrality one Ni^{2+} is oxidized to Ni^{3+}.

Such solid acts as P- type of Semi-conductor.

Q. If NaCl is doped with 10^{-3} mole % of $SrCl_2$, find the concentration of Cation Vacancies.

Sol^n 10^{-5} moles of Sr^{2+} are doped in one mole of NaCl.

No. of Vacancies generated = no. of Sr^{2+} ions doped.

$= 6.022 \times 10^{23} \times 10^{-5}$ per mole of NaCl.

$= 6.022 \times 10^{18}$ Vacancies per mole of NaCl.

ii) **Interstitial impurity defects:**

This arise when small sized atoms of elements (like B, C, N, H etc) are doped into molten metals. The small atom occupy the void (or interstitial) space and improve mechanical & electrical property.

eg:- 'C' is added in iron to increase hardness strength of iron.

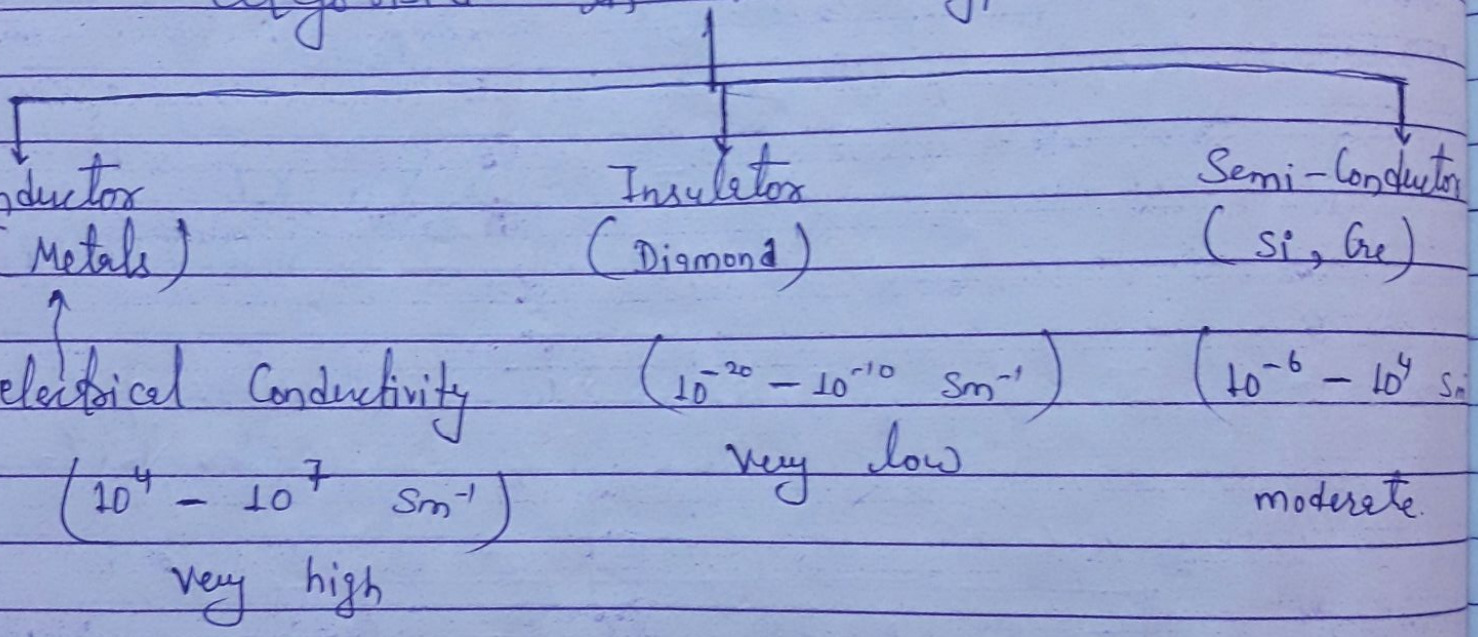
- B, As, Ga are doped in Ge, Si etc.

(22) Properties of solids :-

- electrical properties
- Magnetical properties.

⊕ Electrical Property :-

→ Based on electrical Conductivity, solids are Categorised in three types.



→ Metals with electrical Conductivity of the order of $10^7\ Sm^{-1}$ are very good electrical Conductor (eg:- Cu, Ag, Al etc).

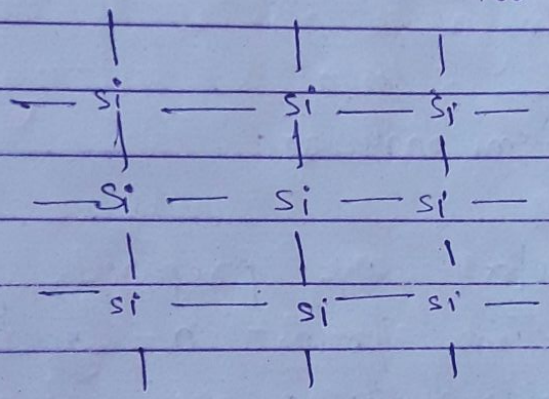
→ According to the band model :- {Read NCERT}

* Pure Silicon, Ge are of no practical used due to very low electrical Conductivity.

To improve Conductivity, of Si, Ge Some impurities are added, called doping.

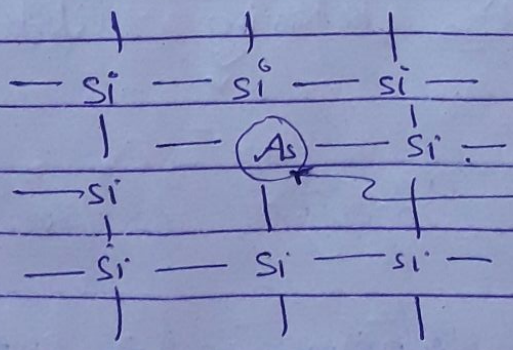
i) Doping of electron rich impurities :-

Group 15 elements eg. P, As etc.



perfect Si Crystal (pure Si).

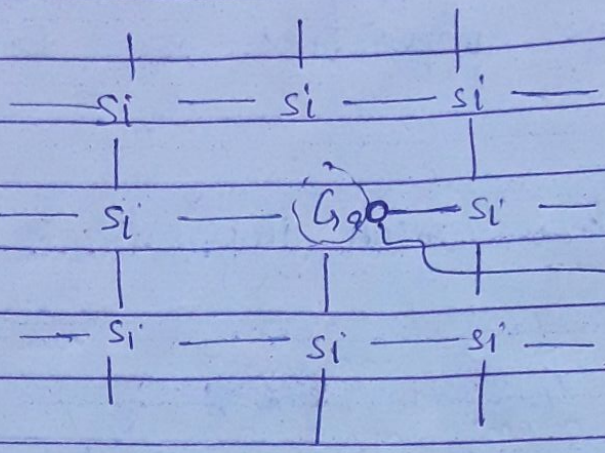
As is doped at some lattice sites



n-type of semiConductor

ii) Doping of electron difficient impurities :-

eg :- B, Al, Ga etc (group 13)



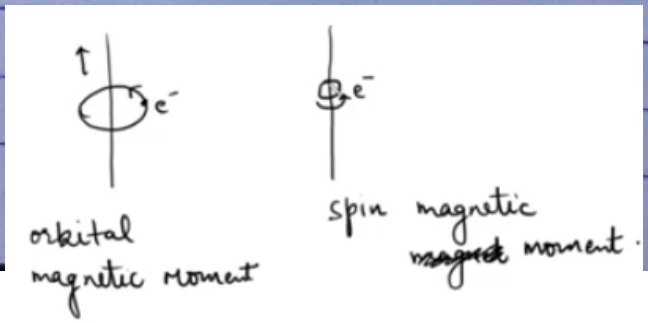
P- type Semiconductor

* Diode & transistors are prepared for electronic application from n-types & p-type of semiconductor.

TiO, CrO_2 & ReO_3 behaving like metals.

(23) Magnetic Property of solids :-

→ Each unpaired electron in a substance acts as a tiny magnet due to its magnetic moment (orbital & spin).



→ For a pair of electron in an orbital, their magnetic moments are cancelled.

→ Compounds (solids) are classified based on their magnetic properties.

i) Diamagnetic substance :-

→ have zero unpaired e^- & are weakely repelled by the applied magnetic field.

eg:- NaCl, H_2O, C_6H_6, $CaCl_2$, diamond etc.

ii) Paramagnetic substance :-

→ have one or more unpaired electrons.

→ Such substance are weakly attracts by magnetic fields.

eg:- O_2, Cr^{3+}, Cu^{2+}, NO etc.

iii) Ferromagnetic substance :-

used for making permanent magnets

→ In metals like Fe, Co, Ni, Gd, CrO_2, metal ions are grouped into region called domains.

→ These domains are randomly oriented in absence of any external magnetic field. & their magnetic moments are cancelled.

→ Under magnetic field, all domains of ferromagnetic substance are oriented in same direction &

produce a very strong magnetic field effect. These metals are permanently magnetised even when mag. field is removed.

eg:- Fe, Co, Ni etc

iv) Anti-ferromagnetic substance :-

→ Equal no. of domains are oppositely oriented in magnetic field & do not show magnetic effect (effect gets Cancelled).

→ eg:- MnO

v) Ferrimagnetic Substance :-

→ Unequal no. of domains are oriented in opposite direction under magnetic field & hence produce weak magnetic effect.

eg:- ~~Fe_3O_4~~, ferrites like

Fe_3O_4 ↓

$MgFe_2O_4$, $ZnFe_2O_4$

* On heating ferrimagnetic substance Converts into paramagnetic substance.